Asterisms, the Zodiac and the Brightest Stars

What you will be able to do

1. Explain the difference between a **constellation** and an **asterism**.
2. Name the 12 zodiac constellations (and the "13th", Ophiuchus) with their rashis.
3. Recall the 20 brightest stars: name, constellation, colour and Indian name.
4. Tell a planet from a star by naked eye.
5. Recognise the major asterisms and constellations visible from India **by their shape**.

How to study this lecture

All sky images here were made with the free planetarium program **Stellarium** for the real sky. The *left* image of each pair shows stick figures and star names; the *right* shows the traditional **constellation art**. North is up and East is to the *left*, as when you look up at the sky. Cover the labels and try to name the stars — your picture quiz will show these patterns unlabelled and even rotated, just as the sky looks at different times of night.

1. Constellation vs asterism

	Constellation	Asterism
Definition	one of the **88 official IAU regions**	any **recognisable star pattern** that is not an official constellation
Example	Ursa Major (the Great Bear)	the **Big Dipper / Saptarishi** — just 7 stars of Ursa Major
Can span several constellations?	no	**yes** — the Summer Triangle uses stars of three constellations

Asterisms are usually **easier to spot** than whole constellations, so we use them as **signposts**.

2. The zodiac: 12 + 1

The zodiac constellations are the ones the Sun, Moon and planets pass through — they lie along the ecliptic.

Rashi	Constellation	Meaning	Brightest star	Sun in it (IAU boundaries, approx.)
Mesha	Aries	ram	Hamal	19 Apr – 13 May
Vrishabha	Taurus	bull	**Aldebaran**	14 May – 20 Jun
Mithuna	Gemini	twins	**Pollux**	21 Jun – 20 Jul

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(continued)

Rashi	Constellation	Meaning	Brightest star	Sun in it (IAU boundaries, approx.)
Karka	Cancer	crab	(faint; Beehive cluster)	21 Jul – 9 Aug
Simha	Leo	lion	**Regulus**	10 Aug – 15 Sep
Kanya	Virgo	maiden	**Spica**	16 Sep – 30 Oct
Tula	Libra	scales	Zubeneschamali	31 Oct – 22 Nov
Vrischika	Scorpius	scorpion	**Antares**	23 – 29 Nov (only ~7 days!)
—	**Ophiuchus**	serpent-bearer	Rasalhague	**30 Nov – 17 Dec**
Dhanu	Sagittarius	archer	Kaus Australis	18 Dec – 18 Jan
Makara	Capricornus	sea-goat	Deneb Algedi	19 Jan – 15 Feb
Kumbha	Aquarius	water-bearer	Sadalsuud	16 Feb – 11 Mar
Meena	Pisces	fishes	(faint)	12 Mar – 18 Apr

Notice: the constellations have **unequal sizes** (the Sun spends ~45 days in Virgo but ~7 in Scorpius); the ecliptic also crosses **Ophiuchus**; and the **rashis** of Indian calendars are *equal* 30° divisions (Lecture 1.2), so their dates differ from these.

3. The brightest stars: your "Top 20"

#	Star	Constellation	Mag.	Colour	Distance (ly)	Indian name
1	**Sirius**	Canis Major	−1.46	white	8.6	Lubdhaka
2	**Canopus**	Carina	−0.74	white-yellow	310	Agastya
3	**Alpha Centauri**	Centaurus	−0.27	yellow	4.37	—
4	**Arcturus**	Boötes	−0.05	orange	37	Swati
5	**Vega**	Lyra	0.03	white	25	Abhijit
6	**Capella**	Auriga	0.08	yellow	43	Brahmahridaya
7	**Rigel**	Orion	0.13	blue-white	~860	—
8	**Procyon**	Canis Minor	0.34	yellow-white	11.5	—
9	Achernar	Eridanus	0.46	blue-white	140	—
10	**Betelgeuse**	Orion	~ 0.5 (var.)	red-orange	~550	Ardra
11	Hadar	Centaurus	0.61	blue-white	~390	—
12	**Altair**	Aquila	0.76	white	17	Shravana
13	Acrux	Crux	0.76	blue-white	~320	—
14	**Aldebaran**	Taurus	0.86	orange	65	Rohini
15	**Antares**	Scorpius	~ 1.0 (var.)	red	~550	Jyeshtha
16	**Spica**	Virgo	0.97	blue-white	250	Chitra
17	**Pollux**	Gemini	1.14	orange	34	Punarvasu
18	Fomalhaut	Piscis Austrinus	1.16	white	25	—
19	**Deneb**	Cygnus	1.25	white	~2,600	—
20	Mimosa	Crux	1.25	blue-white	~280	—
21	**Regulus**	Leo	1.35	blue-white	79	Magha
—	**Polaris**	Ursa Minor	1.98	yellow-white	~430	Dhruva

("Mag." = *magnitude*, the astronomers' brightness scale: *smaller* numbers mean *brighter*. The brightest

stars have magnitudes near 0 or even negative; the faintest stars you can see from a dark site are about magnitude 6.)

Patterns to notice.

- **Colour shows temperature:** blue-white stars (Rigel, Spica) are hottest (10,000–25,000 K); white (Sirius, Vega) ~9,500 K; yellow (the Sun, Capella) ~5,500 K; orange (Arcturus, Aldebaran) ~4,000–4,500 K; red (Betelgeuse, Antares) ~3,500 K.
- **Bright ≠ near:** Deneb looks almost as bright as Pollux but is about 75 times farther away — so it must be enormously more powerful.
- **Polaris is not the brightest star** (about the 48th); it is famous because it hardly moves.
- **Antares** means "rival of Ares (Mars)" — easily mistaken for the red planet when Mars is nearby.

4. Star or planet? Five naked-eye tests

Test	Star	Planet
Twinkling	twinkles, especially near the horizon	shines steadily (usually)
Position	anywhere in the sky	always near the **ecliptic**
Movement	fixed pattern for centuries	moves among the stars over days to weeks
Brightness	brightest is Sirius	Venus, Jupiter and sometimes Mars outshine every star
In binoculars	always a point	Jupiter shows 4 moons; Venus shows phases

Planets twinkle less because they are tiny **discs**, not points — the atmosphere's turbulence averages out across the disc.

5. The signpost patterns: a picture gallery

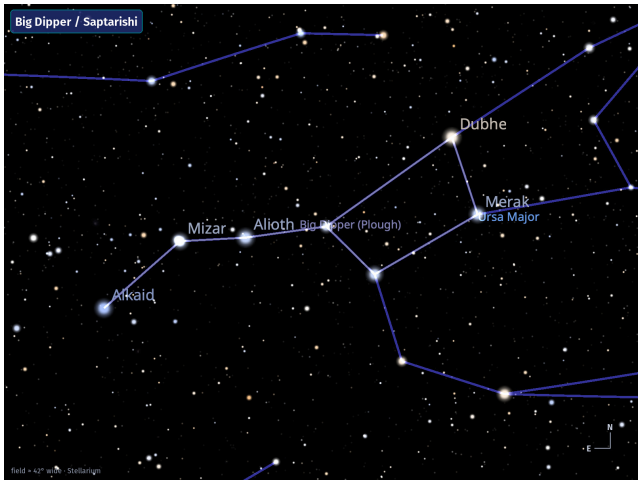
The "W" of Cassiopeia — autumn, northern sky

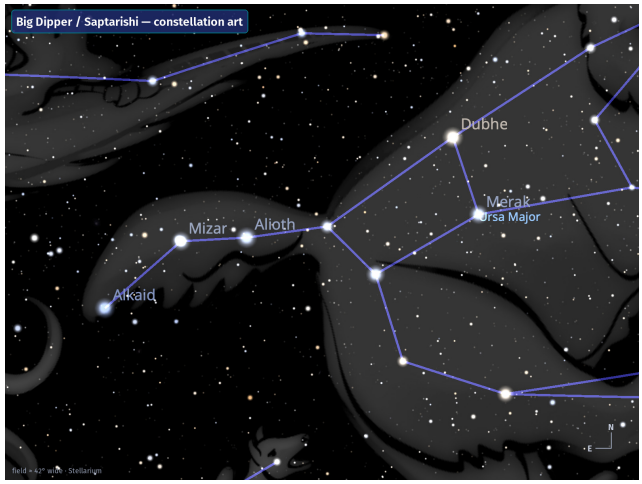




Five stars making a W or M, on the opposite side of Polaris from the Big Dipper. Best in October–December evenings, low in the north.

The Big Dipper (Saptarishi) — spring, northern sky

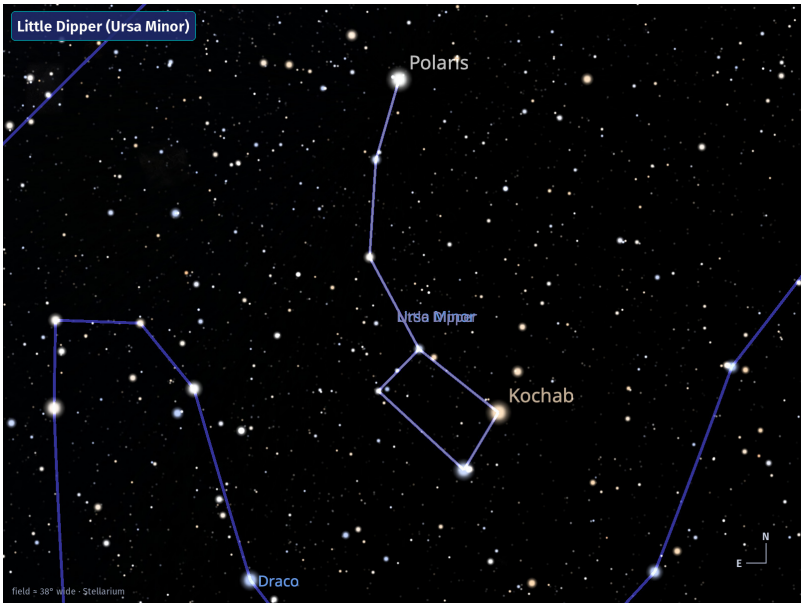




Seven bright stars of Ursa Major form a ladle: 4 in the bowl, 3 in the handle.

- **Pointers:** Merak → Dubhe, extended about 5 times, leads to **Polaris**.
- Follow the curve of the handle onward and you reach the bright orange star **Arcturus**.
- Best from India on evenings of **March–June**, standing above the northern horizon.

The Little Dipper — Ursa Minor, with Polaris



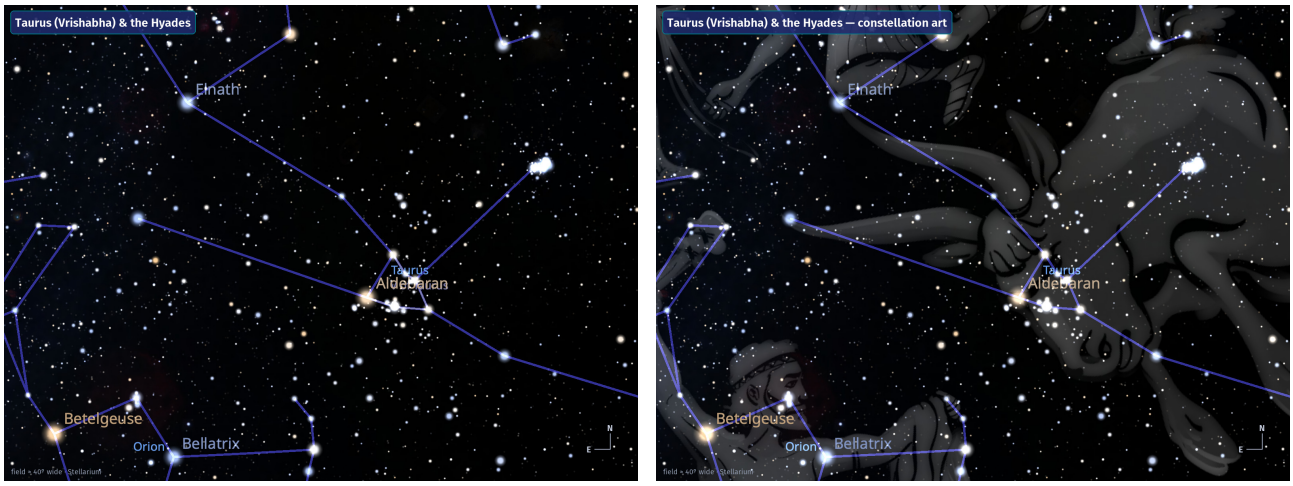
Polaris (Dhruva) is at the end of the handle; Kochab and Pherkad are the “guardians of the pole”. The other stars are faint (magnitude 4–5).

Canis Major and Sirius

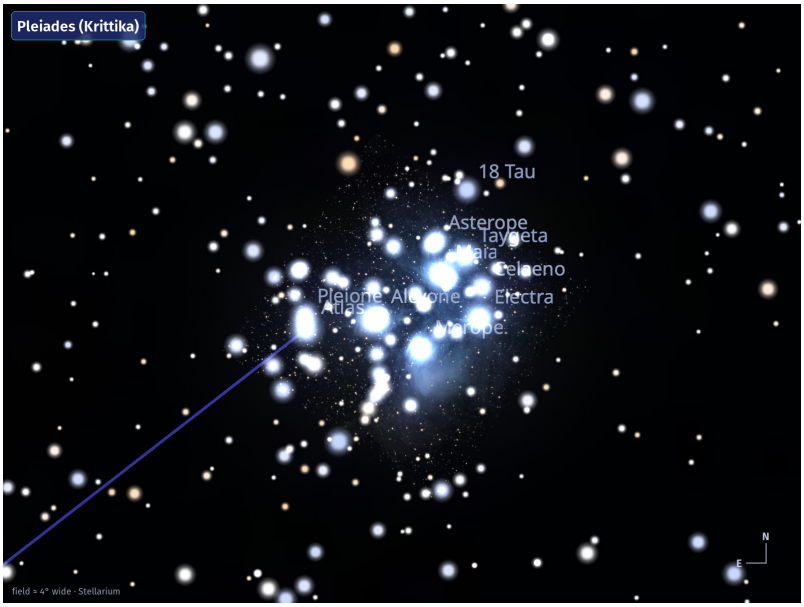


Sirius (*Lubdhaka*), magnitude -1.46, is the brightest star of the night sky, only 8.6 light-years away.

Taurus, the Hyades "V" and the Pleiades



Aldebaran (orange; *Rohini*) marks the bull's eye at the tip of the V-shaped Hyades cluster.



The Pleiades (*Krittika*, the "Seven Sisters"): a true cluster of young blue stars. Most people see 6–7 of them.

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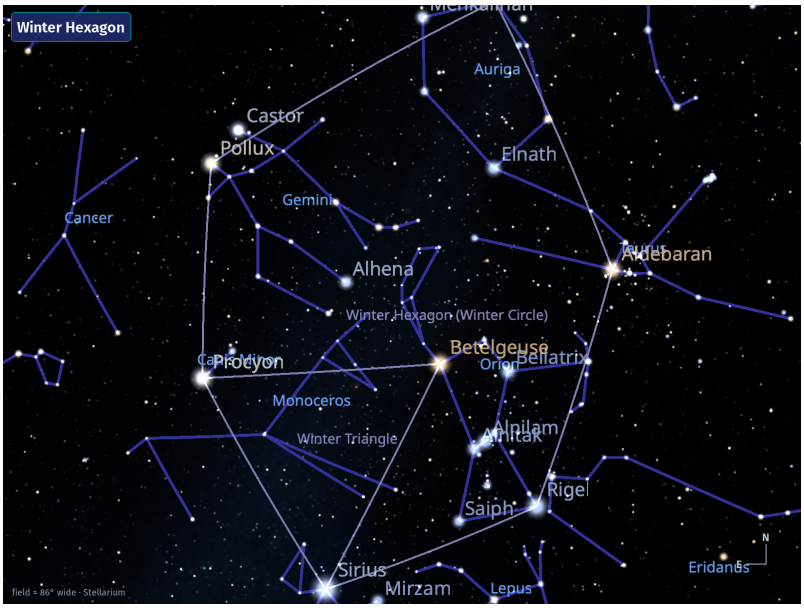
Orion — winter, the king of constellations



Orion, the hunter (*Mriga* in Indian tradition).

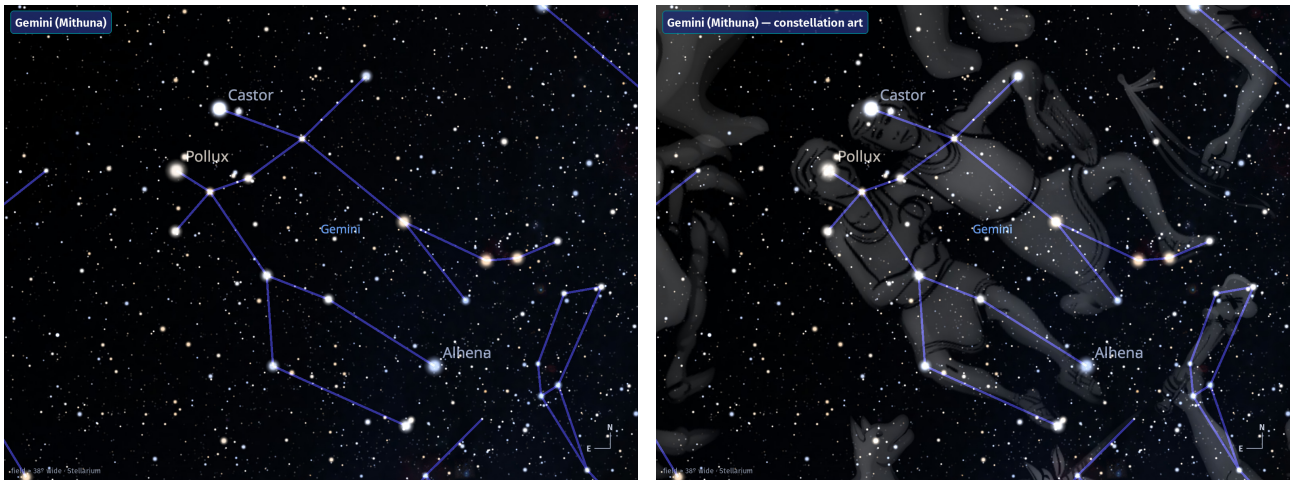
- **Orion's Belt**: three equally bright stars in a line about 3° long — Alnitak, Alnilam, Mintaka.
- **Betelgeuse** (red-orange shoulder; *Ardra*) and **Rigel** (blue-white foot) sit diagonally opposite each other.
- Below the belt hangs the **sword**, with the **Orion Nebula**, a fuzzy patch where stars are being born — visible to the naked eye from a dark site.
- Best in **December–February** evenings, high in the south.

The Winter Hexagon and the Winter Triangle



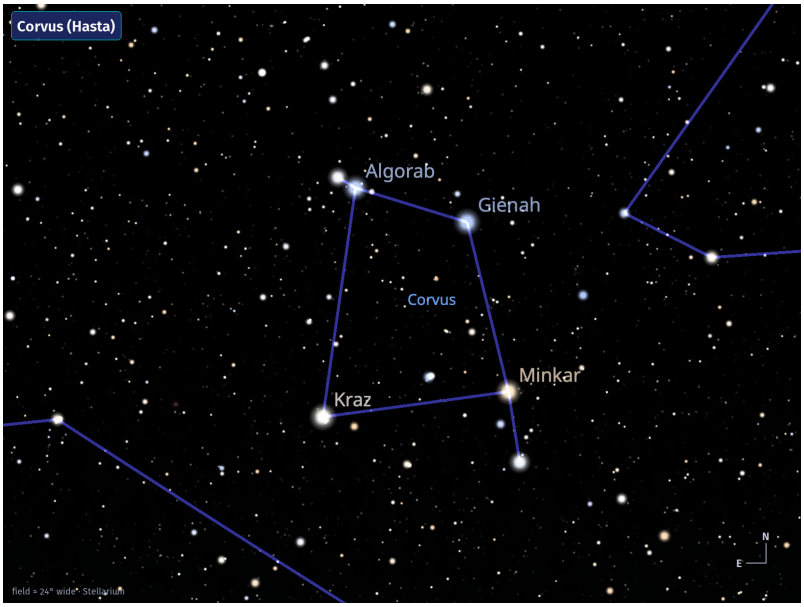
Winter Hexagon: Sirius → Rigel → Aldebaran → Capella → Pollux → Procyon. Winter Triangle: Betelgeuse, Sirius, Procyon.

Gemini — the twins Castor and Pollux (Punarvasu)



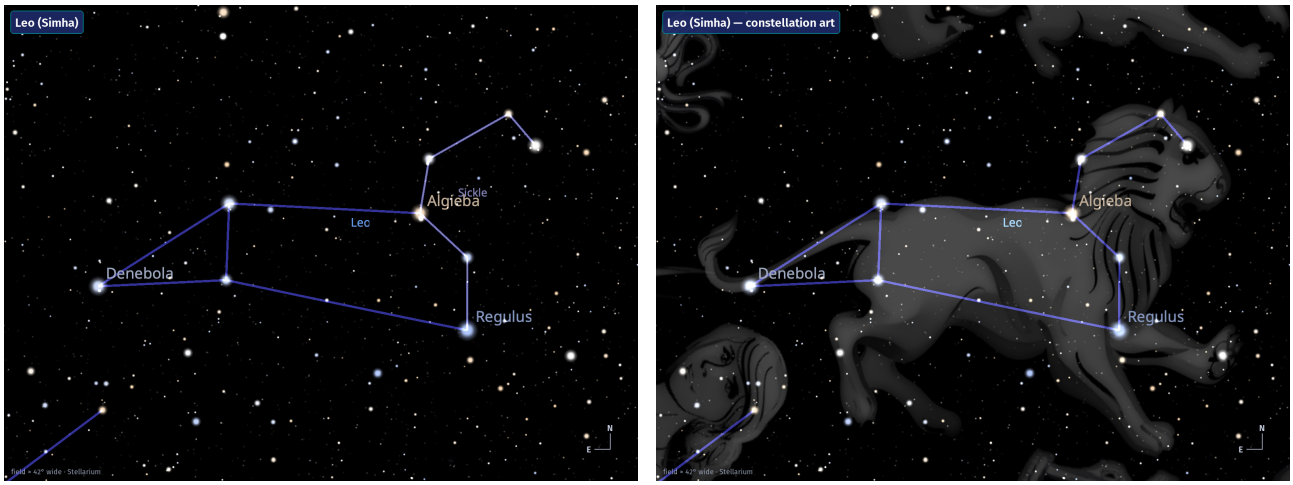
Two bright stars side by side: Pollux (orange, slightly brighter) and Castor (white).

Corvus, the crow (Hasta) — spring



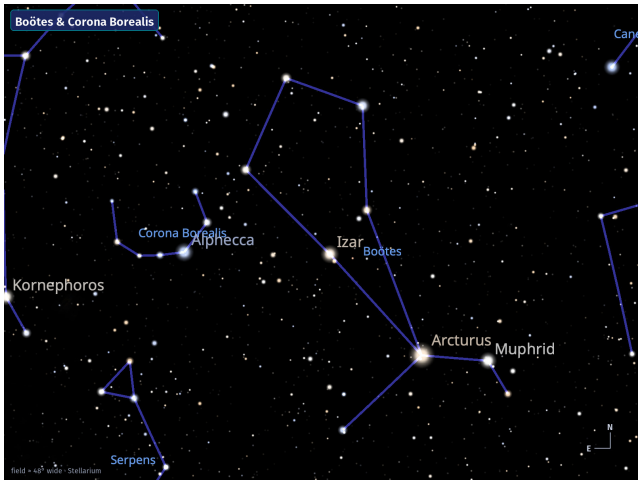
A compact four-star quadrilateral south of Virgo — the *Hasta* (“hand”) nakshatra.

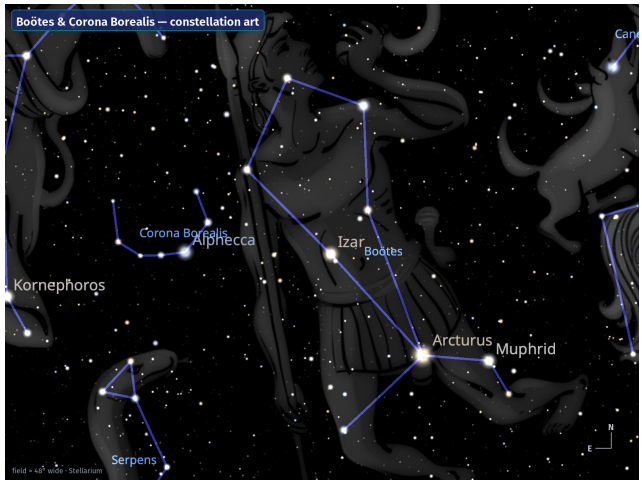
Leo and its Sickle — spring



The Sickle (a backwards question mark) forms the lion’s head and ends in Regulus (*Magha*); Denebola marks the tail.

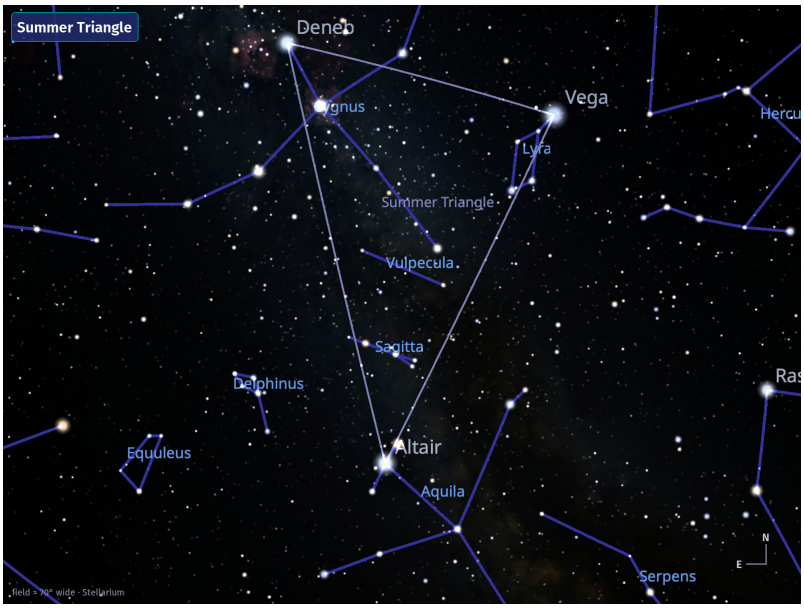
Boötes, the kite, and Corona Borealis



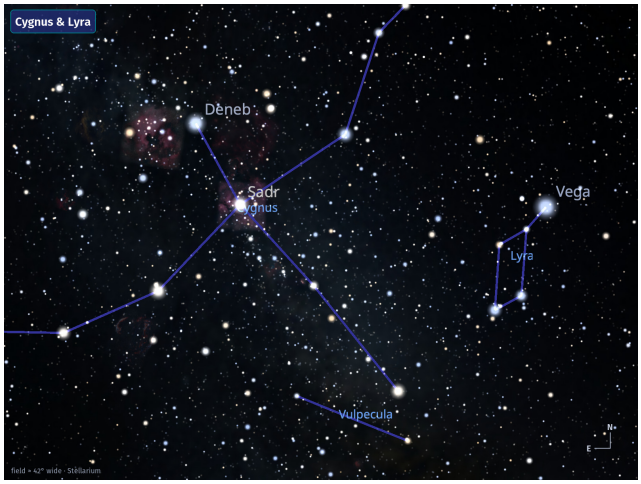


Arcturus (orange; *Swati*) is the brightest star north of the celestial equator. Next door, Corona Borealis is a small, neat semicircle — the Northern Crown.

The Summer Triangle, Cygnus and Lyra



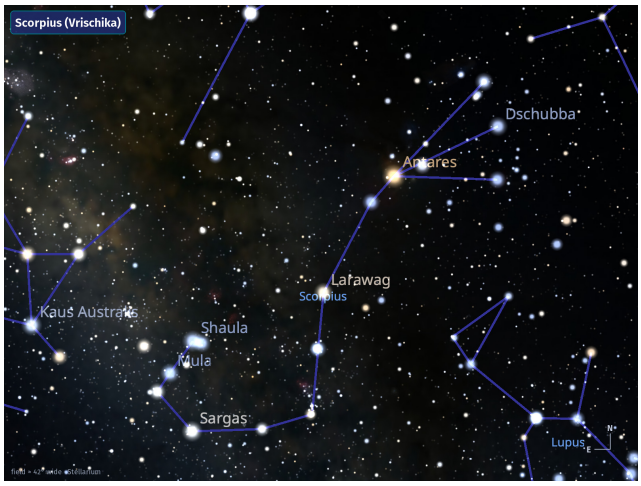
Vega (Lyra; *Abhijit*), Deneb (Cygnus) and Altair (Aquila; *Shravana*) — overhead on evenings from July to November, with the Milky Way running through.

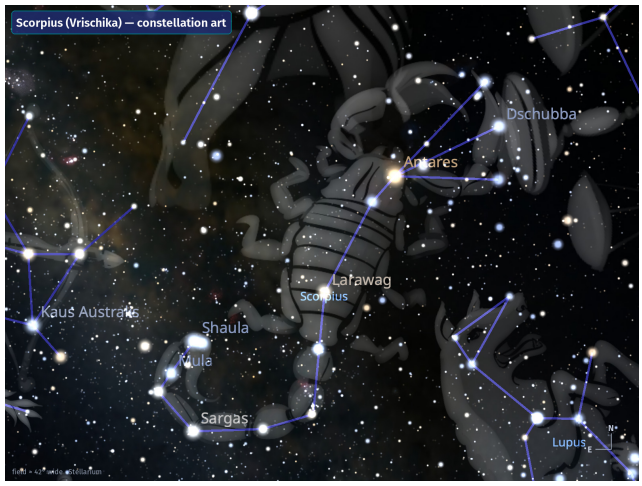




Cygnus forms the **Northern Cross**, with Deneb at the top and Albireo (a gold-and-blue double star in small telescopes) at the foot; Lyra is the small parallelogram hanging from Vega.

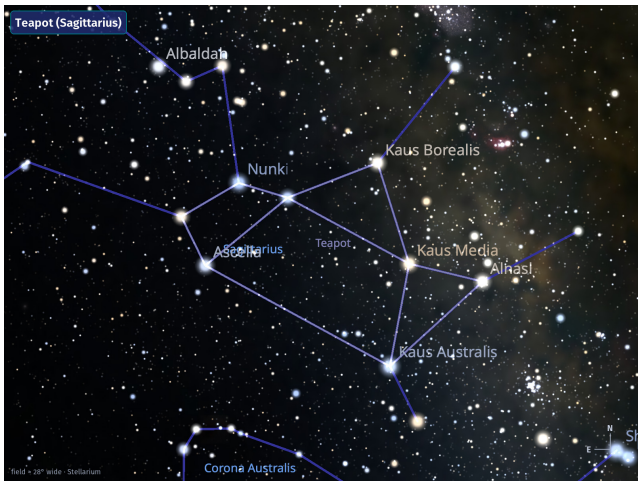
Scorpius — summer and monsoon, southern sky

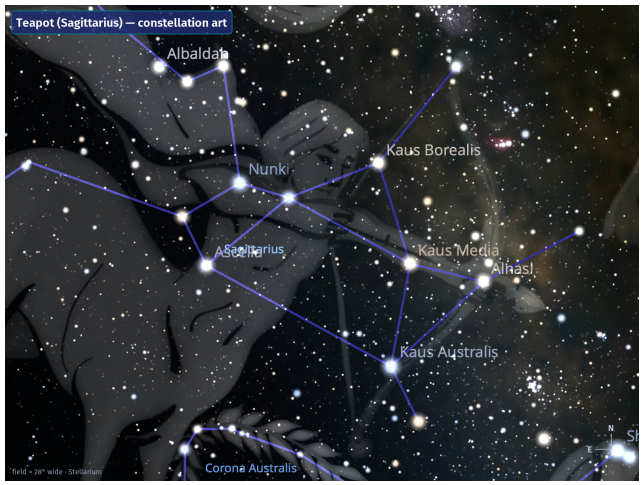




One of the few constellations that looks like its name: a curved tail with a sting and the red heart Antares (*Jyeshtha*). Best May–August evenings, low in the south.

The Teapot (Sagittarius)





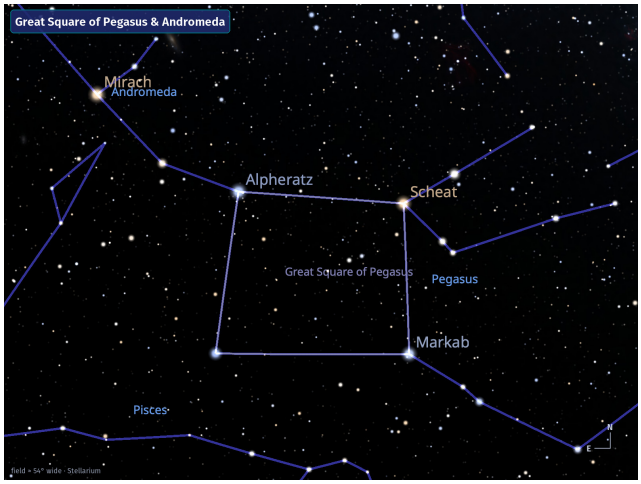
Eight stars of Sagittarius form a teapot. The centre of our Milky Way lies just off the spout, like rising steam.

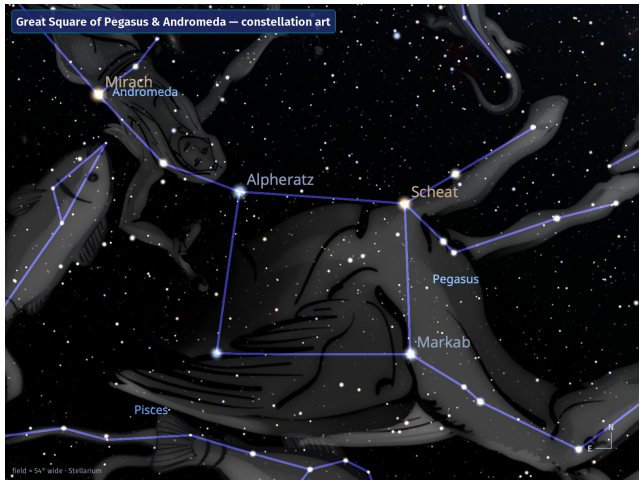
Delphinus (Dhanishtha)



A charming small diamond with a tail, east of Altair — the *Dhanishtha* nakshatra (sometimes pictured as a drum).

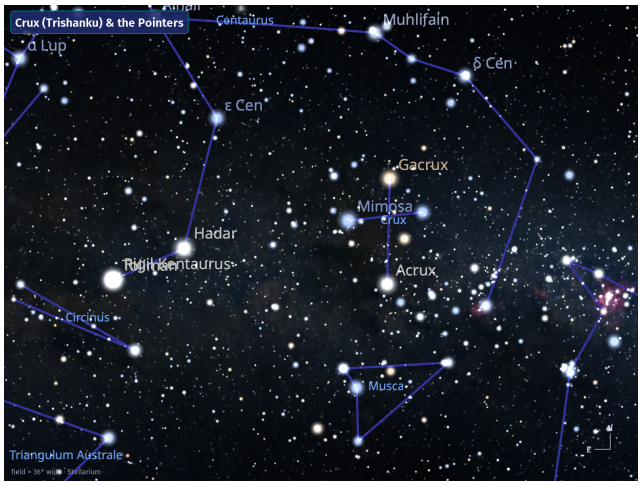
The Great Square of Pegasus and Andromeda — autumn

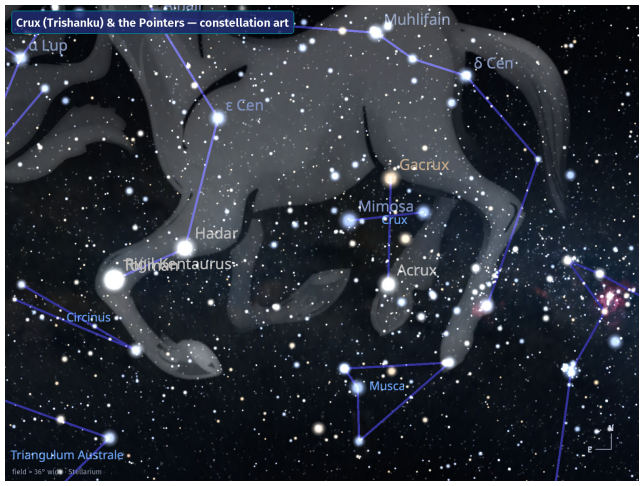




Four similar stars form a square about 15° on a side; one corner, Alpheratz, belongs to Andromeda. From Alpheratz follow the chain to Mirach, then turn up two stars to find the faint smudge of the Andromeda Galaxy.

The Southern Cross (Trishanku) and the Pointers





Crux is the smallest constellation; its long axis points towards the south celestial pole. Alpha Centauri (4.37 light-years) and Hadar are the "Pointers".

Visible from India low in the south on **April–June** evenings, best from south of about 25° N.

Key points to remember

- **Asterism** = unofficial pattern (Big Dipper, Summer Triangle, Teapot, Great Square); **constellation** = official IAU region.
- Signposts: Big Dipper (spring), Orion (winter), Scorpius & Teapot (summer), Summer Triangle (monsoon/autumn), Great Square & Cassiopeia (autumn), Crux (late spring, low south).
- Zodiac: 12 classical constellations + Ophiuchus; rashis are equal 30° divisions.
- Top bright stars: Sirius, Canopus, α Centauri, Arcturus, Vega, Capella, Rigel, Procyon, Achernar, Betelgeuse.
- Colour = temperature; planets twinkle less and stay on the ecliptic.

Check yourself

1. Which three stars form the Summer Triangle, and which constellations do they belong to?
2. Which is the "13th" zodiac constellation?

Check yourself *(continued)*

3. Name two red/orange and two blue-white first-magnitude stars.
4. A bright, steady "star" sits in Leo and has moved since last week. What is it likely to be?

Answers: 1. Vega (Lyra), Deneb (Cygnus), Altair (Aquila). 2. Ophiuchus. 3. E.g. Betelgeuse, Antares (red), Arcturus, Aldebaran (orange); Rigel, Spica (blue-white). 4. A planet — Leo lies on the ecliptic.

Further reading (free / public)

- IAU constellation charts (free PDF maps of all 88) — `https://www.iau.org/public/themes/constellations/`
- OpenStax, *Astronomy 2e*, §2.1 "The Sky Above" — `https://openstax.org/books/astronomy-2e/pages/2-1-the-sky-above`
- H. A. Rey, *The Stars: A New Way to See Them*; G. Consolmagno & D. M. Davis, *Turn Left at Orion* (Cambridge University Press).

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